

1) Convert the following

a) 18_{10}

10010_2

$$\begin{array}{r} 2 \overline{) 18} \quad -0 \\ 9 \quad -1 \\ 2 \overline{) 9} \quad -1 \\ 4 \quad -0 \\ 2 \overline{) 4} \quad -0 \\ 2 \quad -0 \\ 1 \quad -1 \end{array}$$

b) 45_{10}

101101_2

$$\begin{array}{r} 2 \overline{) 45} \quad -1 \\ 22 \quad -0 \\ 2 \overline{) 22} \quad -0 \\ 11 \quad -1 \\ 2 \overline{) 11} \quad -1 \\ 5 \quad -1 \\ 2 \overline{) 5} \quad -1 \\ 2 \quad -0 \\ 1 \quad -1 \end{array}$$

2) a) 10101_2

$$\begin{array}{r} 16 \quad 8 \quad 4 \quad 2 \quad 1 \\ 1 \quad 0 \quad 1 \quad 0 \quad 1 = 21_{10} \end{array}$$

b) 110011_2

$$\begin{array}{r} 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 \\ 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 = 51_{10} \end{array}$$

3) a) 74_{10}

112_8

11111111

$$\begin{array}{r} 8 \overline{) 74} = 2(2) \\ 8 \overline{) 9} = 1 \\ 1 + 1 = 2 \end{array}$$

PT8

let's

b) 285_{10}

377_8

77

$$\begin{array}{r} 8 \overline{) 285} = 7 \\ 8 \overline{) 31} = 7 \end{array}$$

1103043

4) a) 127_8

1101011

$64 \quad 8 \quad 1$

$1 \quad 2 \quad 7 \quad 087_{10}$

$2 - 101 \quad 8$

$2 - 21 \quad 8$

8E21

let's

b) 345_8

$64 \quad 8 \quad 1$

$3 \quad 4 \quad 501 = 229_{10}$

B

5) a) $1A_{16}$

$16 \quad 1$

$1 \quad A \Rightarrow 26_{10}$

1 at 5E

b) $2F_{16}$

$16 \quad 1$

$2 \quad F$

$= 47_{10}$

1010 0101 1100

$$6) \quad 255_{10} / 8$$

$$i) \quad \text{Binary} \Rightarrow 11111111_2$$

$$ii) \quad \text{Octal} \Rightarrow 377_8$$

$$iii) \quad \text{Hexa} \Rightarrow FF_{16}$$

$$7) \quad 1101011_2$$

$$64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1$$

$$1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1$$

$$\Rightarrow 107_{10}$$

$$\text{Octal} \Rightarrow \underline{\underline{153_8}}$$

$$\begin{array}{r|l} 8 & 107 - 3 \\ 8 & 13 - 5 \\ & 1 - 1 \end{array}$$

$$\begin{array}{cc} 0110 & 1011 \\ 6 & B \end{array}$$

$$\text{Hexa} \Rightarrow \underline{\underline{6B_{16}}}$$

$$8) \quad 3AF_{16}$$

$$\text{Decimal} \Rightarrow$$

$$32 \quad 16 \quad 1$$

$$3 \quad A \quad F$$

$$\Rightarrow 933_{10}$$

$$\text{Binary} \Rightarrow$$

$$\underline{\underline{001110100101_2}}$$